

Corner Cover

Evirir has a $n \times m$ grid. The cell in row i and column j is called (i, j) . The *corners cells* are the cells $(1, 1)$, $(1, m)$, $(n, 1)$, and (n, m) .

Evirir wants to choose a $A \times B$ or $B \times A$ subgrid in the grid such that the subgrid contains **at least** two corner cells. Is this possible?

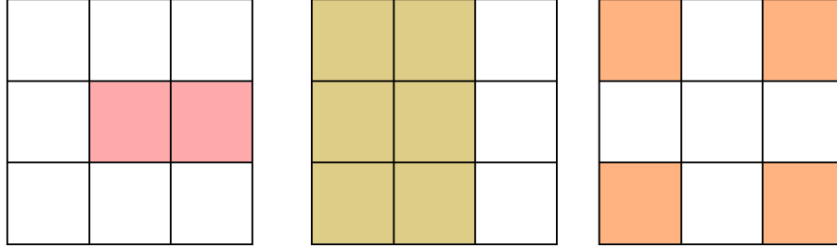


Figure 1: Illustration of two examples of subgrid as well as showing the corner cells

The first two diagrams show some of the possible subgrids (1×2 and 3×2 respectively), and the rightmost diagram shows the 4 corner cells in the grid. The first diagram does not cover two corner cells, but the second one does.

Input Format

There are multiple test cases. The first line contains the number of test cases T . Each of the next T lines contains a test case. The description of a test case follows.

The only line of each test case contains four space-separated integers, n , m , A , and B ($2 \leq n, m, A, B \leq 10^{18}$).

Output Format

Output T lines. For each test case, output YES on a line if it is possible and output NO otherwise.

Constraints

Task	1	2	3	4	5
t	(samples)	1	200	200	200
n, m, A, B	(samples)	10	10	10^9	10^{18}
Points	0	10	10	40	40

Sample Input

3

```

2 3 3 3
2 3 2 3
2 3 2 1

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Sample Output

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NO
YES
YES

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Sample Explanation

In the first test case, it is impossible to choose a 3×3 subgrid from a 2×3 .

In the second test case, it is possible by choosing the entire grid.

In the third test case, one can cover 2 corner cells with a 2×1 subgrid as shown on the left. Note that it is also allowed to use a 1×2 subgrid as shown on the right, but the right subgrid only covers one corner cell and not at least two.

